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Discovery of a 4-aryloxy-1*H*-pyrrolo[3,2-*c*]pyridine and a 1-aryloxyisoquinoline series of TRPA1 antagonists

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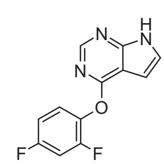
A B S T R A C T

A series of TRPA1 antagonists is described having a 4-aryloxy-1*H*-pyrrolo[3,2-*c*]pyridine or a 1-aryloxyisoquinoline scaffold. These compounds have high ligand efficiency and favorable physical properties and may thus serve as scaffolds for further optimization.

The Transient Receptor Potential A1 (TRPA1) ion channel serves to respond to a variety of sensory stimuli. For example, it mediates responses to exogenous chemical irritants, endogenous inflammatory mediators, zinc, voltage, temperature and mechano-sensitization such as stretch. TRPA1 has been investigated as a target for various potential therapeutic applications including neuropathic and inflammatory pain,[1–3] airway disorders,[4,5] and in bladder oversensitization.[6,7] In particular, Glenmark has initiated Phase-2 clinical trials with GRC-17536 in two studies, one for neuropathic pain and another for asthma. Hydra/Cubist has advanced CB-189625 into Phase-1 for nociceptive pain. Various groups have recently explored different chemical series as TRPA1 antagonists[8–10] and this area has been described in several excellent reviews.[11–15] Several classes of TRPA1 antagonists have been demonstrated, but they typically have one or more of the following liabilities: high lipophilicity, high molecular weight and/or low solubility. Here we describe a novel series of TRPA1 antagonists with high ligand efficiency and improved physicochemical properties.



1
IC_{50} 0.71 μM

From high throughput screening we identified hit **1**. This compound had good starting potency (IC_{50} 0.71 μM), excellent ligand efficiency[16] (0.46), and good solubility (60 μM). Follow-up near-neighbor screening of structurally close analogs led to the identification of the mono-fluoro analog **2** which had comparable if slightly higher antagonist potency (IC_{50} 0.41 μM and LE 0.51). Based on these initial results, we surveyed aryl analogs **3**–**10** as shown in Table 1. It was apparent that activity was strongly associated with the 4-fluorophenyl substituent and that alternative 4-substitutents led to loss of detectable activity in all cases except where R = H.

To further investigate the role of the phenyl group in **1**, we replaced it with analogs containing saturated cycloalkanes **11**, and **12**, benzyl analog **13**, pyridyl **14** and quinolyl analogs **15** and **16** as shown in Figure 1. These changes all led to complete loss of detectable activity (IC_{50} >32 μM).

Since it was evident that the fluoro-phenyl ring was required, we prepared a subset of compounds in which different positioning and nature of the halogen substituent was explored (Table 2).

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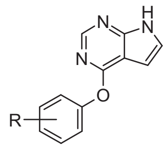
0960-894X/

Table 1
TRPA1 antagonist potency for phenyl-substituted analogs of **1**



Compd	R	IC_{50}[a] (μM)
1	2,4-Bis-F	0.71
2	4-F	0.41
3	H	3.2
4	4-CH_3	>32
5	2-CH_3	>32
6	2,4-Bis-CH_3	>32
7	4-Cl	>32
8	4-CN	>32
9	4-NO_2	>32
10	4-OMe	>32

[a] Results are the mean of at least 2 experiments. Experimental errors are within 20% of value based on replicate measurements.

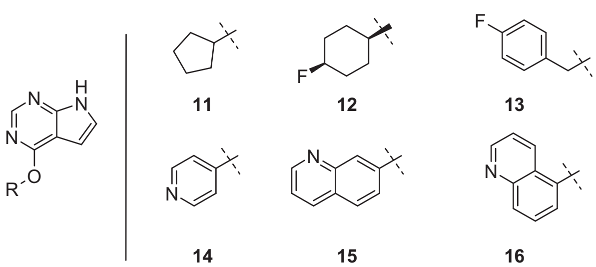
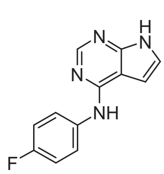


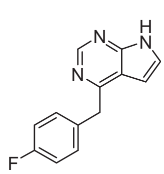
Fig. 1. Analogs containing modifications to the 2-4-bis-fluoroaryl ring of **1**. All these examples were inactive (IC_{50} >32 μM) as TRPA1 antagonists.

Results further showed the strong preference for the 4-F or 2,4-bis-F substitution. Changing to the 2,3-bis-F (**17**) or 3,4-bis-F (**18**) led to moderate reduction in potency. Furthermore, changing the 4-F to 3-F (**20**) or to 4-Cl (**21**) led to loss of all detectable activity (IC_{50} >32 μM).

Modifications to the ether linker also led to loss of activity. In comparison to **2** (IC_{50} 0.41 μM), the nitrogen linked **22** and the methylene linked **23** were both inactive (IC_{50} >32 μM).



22

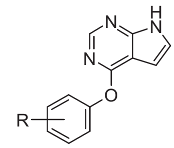


23

Next we evaluated effects of substitution at the pyrrole region of the pyrrolo-pyrimidine core as shown in Table 3. Methylation at position R^1 (affording **24**) and various modifications at R^2 were all relatively well-tolerated and resulted in only moderate reductions in antagonist potency. In comparison to the region occupied by the fluorophenyl substituent, which is intolerant to substitution, the pyrrole region can be modified to a greater extent; these results suggest that the pyrrole region is directed into a larger accessible pocket or toward solvent.

Modifications of the heteroaryl core were evaluated with the analogs shown in Table 4. Elimination of the pyrrole portion of **1** afforded **30**, which was inactive. N-alkylation of the pyrrole nitrogen (**31**), shifting the position of the pyrrole nitrogen (**32**), or

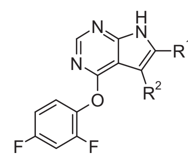
Table 2
Antagonist potency for phenyl-substituted analogs of **1**



Compd	R	IC_{50}[a] (μM)
17	2,3-Bis-F	2.7
18	3,4-Bis-F	3.0
19	2-F	>32
20	3-F	>32
21	4-Cl	>32

[a] Results are the mean of at least 2 experiments. Experimental errors are within 20% of value based on replicate measurements.

Table 3
Antagonist potency for pyrrolo-substituted analogs of **1**

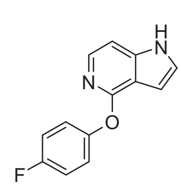


Compd	R^1	R^2	IC_{50}[a] (μM)
22			>32
23			>32
24	Me	H	1.2
25	Me	Me	2.0
26	H	Me	0.65
27	H	Et	1.6
28	H	Cl	0.63
29	H	I	2.5

[a] Results are the mean of at least 2 experiments. Experimental errors are within 20% of value based on replicate measurements.

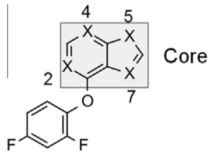
replacing the nitrogen with oxygen (**33** and **34**) all led to loss of detectable activity. Interestingly, the pyrimidine region of the heterocyclic core in **1** appeared more tolerant to modification. Replacement of both nitrogens with carbon at the 2- and 4-positions of **1** afforded **35** and resulted in maintaining nearly equivalent potency (IC_{50} 1.1 μM). Adding back the nitrogen at the 4-position (compound **36**) resulted in loss of detectable activity, whereas adding back the nitrogen at the 2-position (compound **37**) led to an apparent slight increase in potency (IC_{50} 0.32 μM) relative to **1**. These results suggested that the nitrogen at the 4-position was detrimental when compared to the nitrogen at the 2-position. To further assess this, we used **33** (IC_{50} >32 μM) as a reference and prepared the C4 analog, **38** (IC_{50} 1.1 μM). Thus, replacement of N4 with C4 again resulted in increased potency. The same effect is seen upon changing from **34** (IC_{50} >32 μM) to **39** (IC_{50} 2.2 μM).

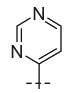
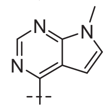
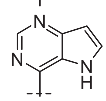
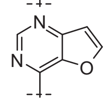
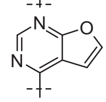
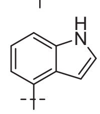
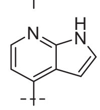
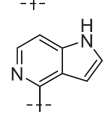
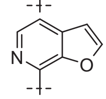
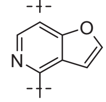
Overall, **37** remained the most potent example to emerge. We had established that the 4-F analog **2** had similar or slightly higher potency than the 2,4-bis-F analog **1**, and therefore we prepared the corresponding modification to **37**; this afforded **40** which again maintained antagonist potency while also reducing molecular weight.



40
IC_{50} 0.31 uM

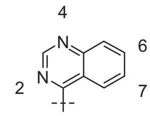
Table 4
Antagonist potency for hetero-modified analogs of **1**



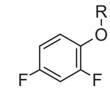
Compd	Core	IC_{50}[a] (μM)
30		>32
31		>32
32		>32
33		>32
34		>32
35		1.1
36		>32
37		0.32
38		1.1
39		2.2

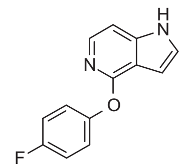
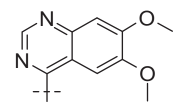
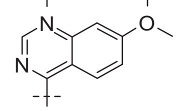
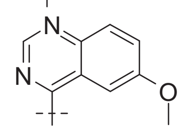
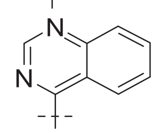
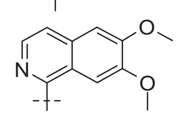
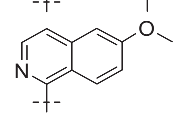
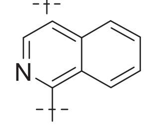
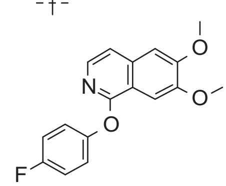
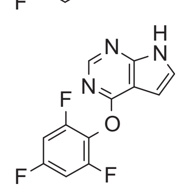
[a] Results are the mean of at least 2 experiments. Experimental errors are within 20% of value based on replicate measurements.

Table 5
Antagonist potency for substituted and core-modified analogs of **1**



Relative positon numbering



Compd	R (or full structure)	IC_{50}[a] (μM)
40		0.31
41		1.2
42		3.0
43		>32
44		>32
45		0.63
46		1.3
47		3.1
48		0.33
49		>32

[a] Results are the mean of at least 2 experiments. Experimental errors are within 20% of value based on replicate measurements.

In parallel with the efforts above, we explored other heterocyclic variations. Near-neighbor mining of our company collection led to the identification of the quinazoline-containing analog **41** (IC_{50} 1.2 μM) which was nearly equipotent with **1**. Comparison among related 6- and 7-OMe analogs indicated that removal of the 7-OMe was tolerated (**42**), whereas removal of the 6-OMe (**43**) or removal of both the 6- and 7-OMe (**44**) were disfavored.

We used this information, along with recognition that changing 4N → 4C in compounds related to **1**, to design the corresponding 4N → 4C analogs **45**, **46**, and **47** (Table 5). From the results above we established that for the pyrrolo-pyridine-related scaffold, replacement of N4 to C4 led to improved potency (see **1** → **37**, **33** → **38**, **34** → **39**); based on this we prepared **45** (IC_{50} 0.63 μM) which had similar potency relative to **41**, as anticipated. Analogous changes from 4N → 4C had similar effect, affording **42** → **46** (IC_{50} 3.0 → 1.3 μM) and **44** → **47** (IC_{50} >32 → 3.1 μM).

Compound **45** was the most potent example to emerge from this series; consequently, we prepared the corresponding 4-fluorophenyl analog.

As described above, activity was improved upon changing the 2,4-bis-fluoroaryl to 4-fluoroaryl (**1** → **2**; IC_{50} 0.71 → 0.41 μM). The same modifications had similar affect upon changing **45** to **48** (IC_{50} 0.63 → 0.33 μM).

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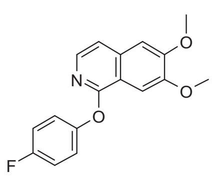
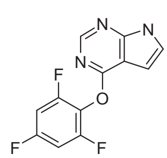
Table 6
Physical and DMPK properties of selected compounds

Compd	**2**	**37**	**40**	**48**
hTRPA1 IC_{50} (μM)	0.41	0.32	0.31	0.33
rTRPA1 IC_{50} (μM)	3.0	1.7	1.2	1.8
hTRPV1 (μM)	>10	>10	nd[i]	nd[i]
LE[16]	0.51	0.49	0.52	0.40
ClogP	2.6	3.4	3.4	4.3
Solubility[a] (μM)	33	77	24	8
Aq. stability[b], pH 1 (% remaining)	10	100	nd[i]	nd[i]
Aq. stability[b], pH 7 (% remaining)	100	100	nd[i]	nd[i]
Aq. Stability[b], pH 10 (% remaining)	100	100	nd[i]	nd[i]
hCLint[c] (μL/min/mg)	24	45	74	110
Human PPB[d] (% free)	11	6	9	2
Caco-2[e] (1E-6 cm/s)	18	33	12	6
Competitive CYP inhibition[f], IC_{50} (μM)	>20	>20	>20	1.9 (1A2)
P450 TDI[g],% shift (isoform)	44 (1A2)	42 (1A2) 23 (3A4)	57 (1A2) 31 (2D6)	No TDI
GSH adduct formation in HLM[h]	+18	+16, +18	+16	+2

[a] Thermodynamic solubility measured in a pH 7.4 buffer using HPLC with UV detection for quantification.
[b] Determined by incubation in buffer at 70 °C for 24 h with monitoring by HPLC with UV detection to determine loss of parent over 24 h.
[c] Metabolic stability performed in human liver microsomes (1 mg/mL) in presence of NADPH using test compound at 1 μM.
[d] Protein binding of test compound at 10 μM in human plasma, as determined by equilibrium dialysis.
[e] Transwell assay. Apparent permeability (apical to basolateral) across a Caco-2 cell monolayer measured at pH 7.4 using test compound at 10 μM applied on the apical side.
[f] Competitive CYP450 (1A2, 2C9, 2C19, 2D6, 3A4) inhibition screening assay in human liver microsomes, using standard probe substrates and detected by LC–MS/MS.
[g] CYP450 (1A2, 2C9, 2D6, 3A4) time-dependent inhibition IC_{50} shift screening assay in human liver microsomes at 10 μM test compound. TDI risk is considered significant in this assay if % shift is >20%; data for compounds that meet this criterion are shown.
[h] GSH adduct formation in human liver microsomes supplemented with 5 mM GSH, as detected by MS/MS using precursor ion scan. Result shows the molecular weight added to the parent-glutathione adduct of the primary metabolite.
[i] Not determined.



48
IC50 0.33 μM



49
IC50 >32 μM

The most potent compounds to emerge were further evaluated and the results are summarized in Table 6. Their antagonist potencies range from IC_{50} 0.31 to 0.41 μM, and all have particularly high ligand efficiency (⩾0.40), moderate solubility (except for **48**), and low lipophilicity (except for **48**). To assess selectivity we tested many compounds against TRPV1; all compounds tested across this series appeared inactive against TRPV1 (IC_{50} >10 μM). Species-specific differences are well known for TRPA1; antagonists that are potent at the human channel are typically much weaker at the mouse and/or rat homologs. This is also true for the series presented here, where we see a 4–7 fold reduction in potency against rat.

A concern for this series is that antagonist activity could potentially be due to pseudo-reversible or irreversible mechanisms. It is known that most TRPA1 agonists are highly electrophilic and react with a thiol cysteine in the channel.[8,14] Thus it is possible that the requisite para-fluorophenyl ring serves to react with nucleophilic groups on the channel. However, several lines of evidence suggest that this is not the case here. First, the compounds are very stable in aqueous solution both under neutral and basic conditions. Second, compound **3** which lacks fluorine substitution) maintains significant, albeit weaker, antagonist potency (IC_{50} 3.2 μM). Furthermore, compound **49**, the tri-F analog of **1** had no detectable activity (IC_{50} >32 μM) despite also being subject to nucelophilic reactivity.

We also considered that nucleophiles could potentially react at the heteroaryl core (with liberation of the phenol); however, we do not detect any released phenol on incubation with glutathione. Also, if such a mechanism was important for the antagonist activity, then we would expect that compounds like **35** would be completely inactive due to lack of a nitrogen adjacent to the phenol linkage. And again, compound **49** has the same electrophilic site as **1** yet is inactive.

Selected compounds in this series have been further characterized for their *in vitro* DMPK properties (Table 6). These compounds have moderate to high levels of plasma protein binding and fast rates of human microsomal clearance. While most compounds are cell permeable (except for **48**), all compounds show potential for metabolic activation since they form detectable GSH adducts upon incubation in human liver microsomes. Furthermore, most compounds show P450 time-dependent inhibition. Further optimization of this series is needed to attain more favorable DMPK properties.

TRPA1 remains a very important therapeutic target for pain and other indications. Most of the chemotype classes that have been described so far are highly lipophilic and have low solubility. In addition, many are particularly selective for the human form and much weaker against mouse and rat, thus complicating *in vivo* translational studies. The series of antagonists described here have high ligand efficiency, improved physical properties, and retain (albeit weakened) activity at the rat channel. Consequently, they may serve as leads for further optimization.

Supplementary data

Supplementary data (representative experimental conditions for the synthesis of compounds (**2**, **37**, **40** and **48**) associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.bmcl.2014.04.045.

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16. LE = Ligand efficiency. Here we define LE = $-RT(\ln(IC_{50}))$/(heavy atom count).